

Radar automatic target recognition using complex high-resolution range profiles

L. Du, H. Liu, Z. Bao and J. Zhang

Abstract: Radar high-resolution range profile (HRRP) has received intensive attention from the radar automatic target recognition (RATR) community. Usually, since the initial phase of a complex HRRP is strongly sensitive to target position variation, only the amplitude information in complex HRRPs is used for RATR, whereas the phase information is discarded. However, the remaining phase information except for initial phases in complex HRRPs may also contain valuable target discriminant information. RATR using complex HRRPs is discussed. The complex HRRPs' feature subspace within each target-aspect sector is extracted via principal component analysis as the corresponding template during the training phase; while in the test phase we decide that a test sample belongs to the feature subspace which has the test sample's minimum reconstruction error approximation. It is shown that the whole process is independent of the initial phases, but exploits the remaining phase information in complex HRRPs. Furthermore, to make the proposed recognition method more practical, a fast time-shift compensation algorithm is proposed. In the recognition experiments based on measured data, the proposed recognition method using complex HRRPs achieves better recognition results than that using only the amplitude vectors of the complex HRRPs.

1 Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centres onto the radar line-of-sight (LOS). It contains target structure signatures such as target size, scatterer distribution and so on, thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1–17]. In recent literatures the properties of target HRRP were discussed [4, 5]; the variation of HRRP's recognition performance with signal bandwidth was studied [6]; the data preprocessing methods for improving the recognition performance of target HRRP was studied [7, 8]; some time-shift invariant features for target HRRP were proposed [9–11]; a new feature selection method for target HRRP was proposed [12]; hidden Markov model was applied to radar HRRP target recognition [13]; radar HRRP-based statistical recognition was studied [14–16] and some advanced classifiers, namely support vector machine and Gabor atom network, were applied to HRRP-based RATR [8, 17]. According to the definition of HRRP given in the earlier literatures, an HRRP is a real vector, which is the amplitude of complex returned echo vector in baseband. Then how the recognition performance of the complex HRRPs containing the phase information will be? However, there has been

little work done in the field of RATR using complex HRRPs. We believe that the reason is complex HRRPs' strong sensitivity to target position variation. Although the literature [18] directly applied the adaptive Gaussian classifier which can be used in real HRRP-based RATR [16] to complex HRRP-based RATR, it did not study the special characteristics of complex HRRPs.

Similar to the scatterer distribution information contained in real HRRPs, the phase information contained in complex HRRPs should also be valuable for RATR. However, the initial phase of a complex HRRP, which is a function of the ratio of target distance to radar wavelength, is very sensitive to the variation of target's radial distance. Assume that a target's returned echo in radio frequency is $s(t - \tau)e^{j\omega_c(t - \tau)}$, where $s(\cdot)$ is the complex envelope, ω_c is the carrier angular frequency and $\tau = 2R/C$ is the time delay with C denoting the speed of light and R representing the radial distance between the target and radar, thereby the returned echo in baseband is $s(t - \tau)e^{-j\omega_c\tau}$, where the phase $\psi_\tau = -\omega_c\tau = -(4\pi/\lambda)R$ with λ denoting the radar wavelength is referred to as the initial phase. Generally, the radar wavelength is so small that the estimated target distance is not accurate enough to compensate the initial phase. For example, for carrier angular frequency of $2\pi \times 5520$ MHz (approximately, the wavelength of 5 cm), if the radial displacement of a target is 2.5 cm, the initial phase difference will be 2π . Therefore how to deal with the initial phase sensitivity of complex HRRPs is a challenging problem in RATR using complex HRRPs. Nevertheless, the literature [18] did not deal with this problem, since the data used in the work of Jacobs [18] are simulated turntable data and targets are assumed to only rotate around the turntable centre in the simulation.

Principal component analysis (PCA) or Karhunen–Loève (KL) transform is a well-known feature extraction method in pattern recognition area [19]. With PCA or KL transform, the extracted feature vectors of a training

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data set $\{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m, \dots, \mathbf{x}_{M-1}\}$ from target T_d ($\mathbf{x}_m$ is a N -dimensional vector) are the eigenvectors corresponding to the p ($p < N$) largest eigenvalues (also referred to as the principal components) of the correlation matrix $\mathbf{R}_x$, which span a p -dimensional feature subspace (referred to as the PCA-based feature subspace), and then if a test sample $\mathbf{x}$ is from target T_d , the projection of $\mathbf{x}$ onto the p -dimensional feature subspace, denoted as $\hat{\mathbf{x}}$, will be its minimum reconstruction error approximation, compared with any other approximation of $\mathbf{x}$ by a p -dimensional vector. This recognition method is referred to as the PCA-based minimum reconstruction error approximation. Since the correlation matrix $\mathbf{R}_x = E[\mathbf{x}_m \cdot \mathbf{x}_m^H] \simeq (1/M) \sum_{m=0}^{M-1} \mathbf{x}_m \cdot \mathbf{x}_m^H$ (with H denoting the conjugate transpose operation) and the reconstruction error $e = E[\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2]$ (with $\|\cdot\|_2$ denoting the L_2 norm operation), for complex HRRPs, the initial phases of training or test samples are counteracted by the outer or inner product operation, however other phase information remains. Thus the PCA-based feature subspace and the minimum reconstruction error approximation, as the corresponding template and the classification algorithm for complex HRRP-based RATR, are completely independent of the initial phases but exploit the remaining phase information in complex HRRPs. In the recognition experiments based on measured data, the PCA-based minimum reconstruction error approximation using complex HRRPs achieves better recognition results than that using real HRRPs.

2 Characteristic comparison between real HRRPs and complex HRRPs

Suppose there are radial displacement and uniform rotation for a target. If the transmitted signal is $s(t)e^{j\omega_c t}$, the m th complex returned echo in the n th range cell ($n = 1, 2, \dots, N$) in baseband is

$$x_n(t, m) = \sum_{i=1}^{L_n} \sigma_{ni} s\left(t - \frac{2R_{ni}(m)}{C}\right) e^{-j[(4\pi/\lambda)R_{ni}(m)]} \quad (1)$$

where L_n is the number of target scatterers in the n th range cell, σ_{ni} is the strength of the i th scatterer of the n th range cell, $R_{ni}(m)$ denotes the radial distance between the radar and the i th scatterer of the n th range cell in the m th returned echo. As the complex envelope $s(\cdot)$ approximates to be unvaried with the radial displacements for all scatterers in one range resolution cell, (1) can be approximated as

$$\begin{aligned} x_n(t, m) &\simeq s(t) \sum_{i=1}^{L_n} \sigma_{ni} e^{-j[(4\pi/\lambda)(R(m) + \Delta r_{ni}(m))]} \\ &= s(t) e^{j\psi(m)} \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)} \end{aligned} \quad (2)$$

where $R(m)$ denotes the radial distance between the target reference centre in the m th returned echo and the radar, $\Delta r_{ni}(m)$ represents the radial displacement of the i th scatterer of the n th range cell in the m th returned echo [if the target-radar orientation changes, $\Delta r_{ni}(m)$ will change accordingly], and $\psi(m) = -(4\pi/\lambda)R(m)$ denotes the initial phase of the m th returned echo. Without loss of generality, we assume that the transmitted signal $s(t)$ is a rectangular pulse with unit

amplitude, then (2) can be written as

$$x_n(m) = e^{j\psi(m)} \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)} \quad (3)$$

2.1 Characteristics of real HRRPs

As the aforementioned derivation, the m th real HRRP can be defined as

$$\begin{aligned} \mathbf{x}(m) &\triangleq [|x_1(m)|, |x_2(m)|, \dots, |x_n(m)|, \dots, |x_N(m)|]^T \\ &= \left[\left| \sum_{i=1}^{L_1} \sigma_{1i} e^{j\phi_{1i}(m)} \right|, \left| \sum_{i=1}^{L_2} \sigma_{2i} e^{j\phi_{2i}(m)} \right|, \dots, \right. \\ &\quad \left. \left| \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)} \right|, \dots, \left| \sum_{i=1}^{L_N} \sigma_{Ni} e^{j\phi_{Ni}(m)} \right| \right]^T \end{aligned} \quad (4)$$

where T denotes the transpose operation. Obviously, real HRRPs are unvaried with the initial phases. As discussed in the literatures [9, 11], it is a prerequisite of real HRRP-based RATR to deal with the target-aspect, time-shift and amplitude-scale sensitivity problems of real HRRP. According to the scattering centre model [5, 20], which has been proved to be a simple and efficient target approximate model (actually a first-order approximation model) in synthetic aperture radar (SAR) imaging and inverse synthetic aperture radar (ISAR) imaging, the scatterer distribution of a target will vary with the target aspect. This is referred to as the target-aspect sensitivity. Usually, a subset of HRRPs from a target-aspect sector without scatterers' motion through range cells (MTRC) is defined to be an HRRP frame, which can be represented by a frame template. The time-shift sensitivity is caused by the move of target signal in range window. There are two kinds of time-shift sensitivity in RATR using real HRRPs: (i) in the training phase, training samples should be time-shift compensated, which can be achieved by using the time-shift compensation techniques in ISAR imaging, such as the slide correlation processing or its modifications; (ii) in the test phase, before a test sample is matched with training samples, its envelope should be aligned with the training samples' envelopes. Obviously, the time-shift compensation techniques in ISAR imaging are not suitable for this process, since the test sample and the training frame may be from different targets. Generally, the optimal time-shift compensation of the test sample matched with the training samples should be obtained according to the corresponding recognition method. Since a real HRRP usually includes many range cells, it requires large computation burden to search the optimal time-shift compensation by shifting the test sample between range cells one by one. Thus a fast time-shift compensation algorithm should be derived for the corresponding recognition method. The amplitude-scale sensitivity comes from the fact that the intensity of a real HRRP is a function of radar transmitting power, radar distance, radar antenna gain, radar receiver gain, radar system losses and so on. Thus real HRRPs measured by different radars or under different conditions will have different amplitude scales. Therefore usually only the waveforms of real HRRPs without amplitude-scale sensitivity can be utilised in real HRRP-based RATR. This is accomplished by amplitude normalisation, for example L_2 normalisation.

Another important characteristic of real HRRPs is that the average profile of a real HRRP frame can reduce the target-aspect sensitivity of real HRRPs to a large degree [5, 9]. As the above derivation, the power of $x_n(m)$ can be written as

$$|x_n(m)|^2 = x_n(m) \times x_n^*(m) \\ = \sum_{i=1}^{L_n} \sigma_{ni}^2 + 2 \sum_{i=2}^{L_n} \sum_{h=1}^{i-1} \sigma_{ni} \sigma_{nh} \cos[\theta_{nih}(m)] \quad (5)$$

where $*$ denotes the complex conjugation operation, and

$$\theta_{nih}(m) = \phi_{ni}(m) - \phi_{nh}(m) \\ = -\frac{4\pi}{\lambda} [\Delta r_{ni}(m) - \Delta r_{nh}(m)] \quad (6)$$

where $\theta_{nih}(m)$ represents the phase difference between the i th scatterer and the h th scatterer of the n th range cell in the m th returned echo. The first term at the right side of (5) is referred to as the scatterer auto-term (SAT) and the second term is referred to as the scatterer cross-term (SCT). In order to reduce the target-aspect sensitivity of real HRRPs, the objective is in fact the reduction of SCTs. Actually, the SCTs in real HRRPs are similar to the ‘coherent speckles’ in SAR image. In SAR imaging, the average processing is used to reduce the ‘coherent speckles’ [20]. Obviously, the training samples in a frame for RATR using real HRRPs are more than the SAR images which can be used to do average processing. Let $\{\mathbf{x}(0), \mathbf{x}(1), \dots, \mathbf{x}(m), \dots, \mathbf{x}(M-1)\}$ denote a real HRRP frame without scatterers’ MTRC. In applications, the average profile of a real HRRP frame is defined as

$$\bar{\mathbf{x}} \triangleq \left[\sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_1(m)|^2}, \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_2(m)|^2}, \dots, \right. \\ \left. \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_n(m)|^2}, \dots, \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_N(m)|^2} \right]^T \quad (7)$$

We now show that the average profile can reduce the SCTs in a real HRRP frame to a large degree. Let $\tilde{\mathbf{x}}(m) = [x_1(m), x_2(m), \dots, x_N(m)]^T$ represent the complex HRRP of the m th time return. Performing the Fourier transformation to each row of the motion compensated complex HRRP frame $\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}(0), \tilde{\mathbf{x}}(1), \dots, \tilde{\mathbf{x}}(M-1)]_{N \times M}$, which corresponds to each range cell, one can obtain the ISAR image $\mathbf{X} = [\mathbf{X}(0), \mathbf{X}(1), \dots, \mathbf{X}(M-1)]_{N \times M}$, where $X_n(k)$ represents the complex strength summation of the scatterers in the resolution gridding (n, k) , which is referred to as an image pixel. According to the energy preservation property of the Fourier transformation, that is the Parseval theorem

$$\sum_{k=0}^{K-1} |X_n(k)|^2 = \sum_{m=0}^{M-1} |x_n(m)|^2 \quad (8)$$

where the term at the left side is also equal to the stable SAT in the n th range cell approximately, and the one at the right side is identical with the average profile after being amplitude normalised, the average profile defined in (7) can represent the stable SAT approximately and reduce the SCTs in the HRRP frame to a large degree. Mathematically speaking, for a data set $\{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_m, \dots, \mathbf{x}_{M-1}\}$, its average

vector satisfies

$$\bar{\mathbf{x}} = \arg \min_{\mathbf{x}} \sum_{m=0}^{M-1} \|\mathbf{x}_m - \mathbf{x}\|_2^2 \quad (9)$$

while its first principal component, that is the principal component corresponding to the largest eigenvalue of the correlation matrix, satisfies

$$\mathbf{u}_1 = \arg \min_{\mathbf{u}} \sum_{m=0}^{M-1} \|\mathbf{x}_m - \cos(\angle(\mathbf{x}_m, \mathbf{u})) \times \mathbf{u}\|_2^2 \\ = \arg \min_{\mathbf{u}} \sum_{m=0}^{M-1} \left\| \mathbf{x}_m - \frac{\langle \mathbf{x}_m, \mathbf{u} \rangle}{\|\mathbf{x}_m\|_2 \times \|\mathbf{u}\|_2} \times \mathbf{u} \right\|_2^2 \quad (10)$$

where $\angle(\cdot, \cdot)$ denotes the angle between two vectors and $\langle \cdot, \cdot \rangle$ denotes the inner product operation between the two vectors. The average vector and the first principal component will be similar, if and only if the data set assembles compactly. A real HRRP frame has this characteristic therefore the first principal component in the correlation matrix, which is referred to as the principal profile in this paper, can also reduce the target-aspect sensitivity of a real HRRP frame. This can be illustrated by Fig. 1, which comes from an HRRP frame in the measured data of ‘Yark-42’ (detailed introduction shown in Section 4.1). Figs. 1a and b compare the principal profile and the corresponding average profile, which show the two profiles are similar. For the sake of clarity, we only give a section of the profiles. Actually, a measured HRRP is a 256-dimensional vector. As is well known, the direction indicated by the first principal component of a correlation matrix has the largest energy in the feature space therefore the first principal component can represent the most important signature of the data set. From the above analysis, the average profile or the principal profile of a real HRRP frame can be used as the corresponding frame template for real HRRPs.

2.2 Characteristics of complex HRRPs

As the above derivation, the m th complex HRRP is

$$\tilde{\mathbf{x}}(m) = [x_1(m), x_2(m), \dots, x_n(m), \dots, x_N(m)]^T \\ = e^{j\psi(m)} \left[\sum_{i=1}^{L_1} \sigma_{1i} e^{j\phi_{1i}(m)}, \sum_{i=1}^{L_2} \sigma_{2i} e^{j\phi_{2i}(m)}, \dots, \right. \\ \left. \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)}, \dots, \sum_{i=1}^{L_N} \sigma_{Ni} e^{j\phi_{Ni}(m)} \right]^T \quad (11)$$

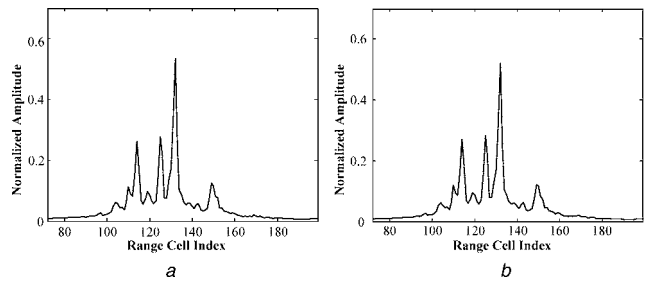


Fig. 1 Principal profile and the average profile of a real HRRP frame in ‘Yark-42’ measured data

The two profiles are similar, since the samples in a real HRRP frame assemble compactly
a Principal profile
b Average profile

Obviously, different from real HRRPs, complex HRRPs vary with initial phases. Since the amplitude profile of a complex HRRP is the corresponding real HRRP, complex HRRPs are also sensitive to the variation of target aspect, time shift and amplitude scale. The approaches to deal with the three sensitivity problems of real HRRPs are also suitable for complex HRRP-based RATR. As discussed in Sections 1 and 2.1, there is another sensitivity problem for complex HRRPs, that is the initial phase sensitivity. Similar to the time-shift sensitivity, there are also two kinds of initial phase sensitivity in RATR using complex HRRPs: (i) in the training phase, the initial phases of each training frame should be calibrated, which can be achieved by using the autofocus techniques in ISAR imaging, such as dominant scatterer algorithm or its modifications; (ii) in the test phase, before a test sample is matched with a training frame, its initial phase should be calibrated with the training samples' initial phases. Obviously, the autofocus techniques in ISAR imaging are not suitable for this process, since the test sample and the training frame may be from different targets. Actually, it is hard to deal with the initial phase sensitivity of complex HRRPs in the test phase of complex HRRP-based RATR.

As discussed in Section 2.1, because the average profile of a real HRRP frame can represent the stable SAT profile, the average profile of each frame can be used as the corresponding frame template in real HRRP-based RATR. Then can the average profile of a complex HRRP frame be used as the corresponding frame template in complex HRRP-based RATR? Actually, because the ISAR image is the Fourier transformation of the complex HRRP frame along the range cell dimension (as discussed in Section 2.1), after being amplitude normalised, the zero-Doppler slice in the ISAR image is identical with the average profile of a motion compensated complex HRRP frame. The average profile of a real HRRP frame can represent the projection of the target's scatterer distribution onto the radial range dimension, whereas the average complex profile of a complex HRRP frame only represents the zero-Doppler slice of the target's scatterer distribution. Therefore the average complex profile cannot be used as the corresponding frame template for complex HRRP-based RATR. Moreover, because the focus location in an ISAR image varies with the autofocus approach, the average complex profile of a complex HRRP frame is not exclusive. This further demonstrates that the average complex profile of a complex HRRP frame is not the suitable frame template for RATR using complex HRRPs.

Therefore for complex HRRPs, the frame template and the classification algorithm should be unvaried with the initial phases of complex HRRPs.

3 Frame template and classification algorithm for complex HRRPs

As discussed in Section 1, the initial phases of a complex HRRP frame are counteracted in the corresponding correlation matrix by outer product operation and those of test samples are also counteracted in the reconstruction error by inner product operation therefore the PCA-based feature subspace of a complex HRRP frame and the minimum reconstruction error approximation can be used as the corresponding frame template and the classification algorithm for complex HRRPs, respectively.

3.1 PCA-based feature space of complex HRRPs

As discussed in Section 2.1, the principal profile of a real HRRP frame can also be used as the corresponding frame template in real HRRP-based RATR. Then can the principal profile of a complex HRRP frame, which is referred to as the principal complex profile in this paper, be used as the corresponding frame template in complex HRRP-based RATR? Shown in Fig. 2a is the ISAR image from the HRRPs in the same frame as that used in Fig. 1. If the ISAR image has one dominant scatterer, the energy of the Doppler slice including the dominant scatterer will be much larger than any other Doppler slice. Thereby, the Doppler slice, which is referred to as the dominant Doppler slice in this paper, should be similar to the corresponding principal complex profile. To demonstrate this deduction, shown in Figs. 2b and c are the amplitude profiles of the principal complex profile and the dominant Doppler slice of the ISAR image. Obviously, the waveforms of the two profiles are similar. Nevertheless, the prerequisite of this deduction is that the ISAR image has one dominant scatterer. If an ISAR image has two or more dominant scatterers, the principal complex profile should be similar to the combination of these dominant Doppler slices.

According to the above analysis, since the principal complex profile of a complex HRRP frame cannot describe the whole information of the target's scatterer distribution, but represents the dominant Doppler slice or the combination of several dominant Doppler slices of the target's scatterer distribution, the principal complex profile, similar to the average complex profile, is not the suitable frame template for RATR using complex HRRPs. Then can we select not only the principal complex profile of a complex HRRP frame but also some other principal components of the correlation matrix, that is the PCA-based feature subspace, as the corresponding frame template for RATR using complex HRRPs? Fig. 3a shows the eigenvalue chart of the complex HRRPs in the same frame as that used in Figs. 1 and 2. It is shown that a 30-dimensional PCA-based feature subspace can approximately hold 99% energy of the whole feature space. Therefore the PCA-based feature subspace of each complex HRRP frame is the suitable frame template for RATR using complex HRRPs. Furthermore, shown in Fig. 3b is the eigenvalue chart of the corresponding real HRRPs for comparison.

3.2 PCA-based minimum reconstruction error approximation

As stated earlier, for complex HRRP-based RATR, the PCA-based feature subspace of a complex HRRP frame can be used as the corresponding frame template in the training phase. Actually, the projection feature vector of a test sample onto the PCA-based feature subspaces of a complex HRRP frame can be applied to many classification algorithms. Nevertheless, in view of the initial phase sensitivity of complex HRRPs in the test phase, we select the minimum reconstruction error approximation as the classification algorithm. The main procedure of the PCA-based minimum squared error approximation in RATR using complex HRRPs will be described in the following.

Divide the training samples of target T_d ($d = 1, 2, \dots, D$) into HRRP frames $\{\chi_{dl}\}_{l=1}^L$, where $\chi_{dl} = \{\mathbf{x}_{dlm}\}_{m=1, 2, \dots, M}$ denotes the l th range aligned HRRP frame and $\mathbf{x}_{dlm}$ denotes an amplitude normalised N -dimensional

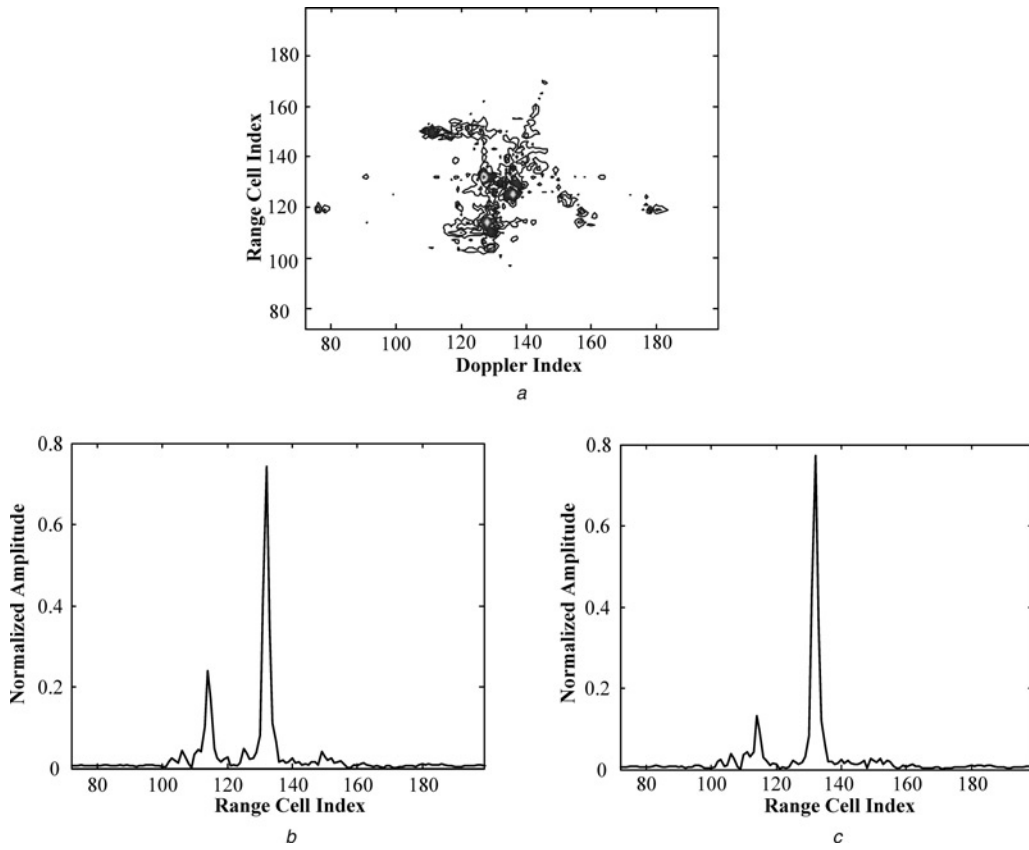


Fig. 2 ISAR image along with its amplitude profiles of the average complex profile and the dominant Doppler slice of a complex HRRP frame in 'Yark-42' measured data

The two amplitude profiles are similar, since the ISAR image has one dominant scatterer

a ISAR image

b Amplitude profile of the principal complex profile

c Amplitude profile of the dominant Doppler slice

complex HRRP. The correlation matrix is

$$\mathbf{R}_{dl} = E\{\mathbf{x}_{dlm} \cdot \mathbf{x}_{dlm}^H\} \simeq \frac{1}{M} \sum_{m=0}^{M-1} \mathbf{x}_{dlm} \cdot \mathbf{x}_{dlm}^H \quad (12)$$

By definition, an eigenvector $\mathbf{u}_{dlq}$ of $\mathbf{R}_{dl}$ satisfies

$$\lambda_{dlq} \mathbf{u}_{dlq} = \mathbf{R}_{dl} \cdot \mathbf{u}_{dlq}, \quad q = 1, 2, \dots, N \quad (13)$$

where λ_{dlq} is the corresponding eigenvalue and $\lambda_{dl1} \geq \lambda_{dl2} \geq \dots \geq \lambda_{dlq} \geq \dots \geq \lambda_{dlN}$. Let η ($0 < \eta < 1$)

denote the threshold, $\eta = 99\%$. If

$$\frac{\sum_{q=1}^p \lambda_{dlq}}{\sum_{q=1}^N \lambda_{dlq}} \geq \eta \quad (14)$$

then the corresponding feature subspace is the space spanned by $\mathbf{U}_{dl} = [\mathbf{u}_{dl1}, \mathbf{u}_{dl2}, \dots, \mathbf{u}_{dlq}, \dots, \mathbf{u}_{dlp}]$. If $\mathbf{x}$ denotes a test sample, namely an amplitude normalised complex HRRP, and $\mathbf{x}_\tau$ denotes its time-shifted vector

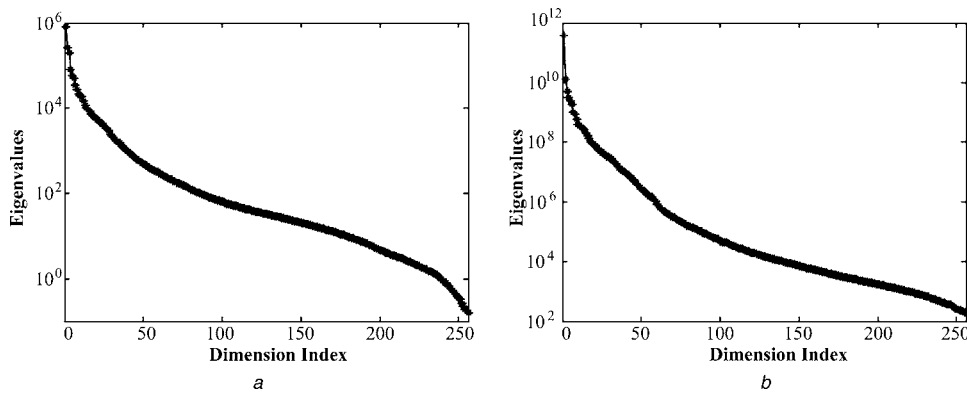


Fig. 3 Eigenvalue charts of the complex HRRPs and the real HRRPs in a frame in 'Yark-42' measured data

Vertical axis is logarithmic

a Eigenvalue chart of the complex HRRPs

b Eigenvalue chart of the real HRRPs

with time shift of τ , by definition, we have

$$\mathbf{x}_\tau = \sum_{q=1}^N \alpha_{dlq|\tau} \mathbf{u}_{dlq} \quad \text{and} \quad \alpha_{dlq|\tau} = \mathbf{u}_{dlq}^H \cdot \mathbf{x}_\tau \quad (15)$$

Then the approximation vector of $\mathbf{x}_\tau$ in the p -dimensional feature subspace $\mathbf{U}_{dl}$ is

$$\begin{aligned} \hat{\mathbf{x}}_{dl|\tau} &= \sum_{q=1}^p \alpha_{dlq|\tau} \mathbf{u}_{dlq} = \sum_{q=1}^p (\mathbf{u}_{dlq}^H \cdot \mathbf{x}_\tau) \mathbf{u}_{dlq} \\ &= \mathbf{U}_{dl} \cdot (\mathbf{U}_{dl}^H \cdot \mathbf{x}_\tau) \end{aligned} \quad (16)$$

The approximation vector with the optimal time-shift compensation is

$$\begin{aligned} \hat{\mathbf{x}}_{dl|\tau^{(o)}} &= \arg \min_{\tau} e_{dl|\tau} \\ &= \arg \min_{\tau} \|\mathbf{x}_\tau - \mathbf{U}_{dl} \cdot (\mathbf{U}_{dl}^H \cdot \mathbf{x}_\tau)\|_2^2 \end{aligned} \quad (17)$$

The minimum reconstruction error among the errors between $\mathbf{x}$ and its approximation vectors in all of the p -dimensional feature subspaces of target T_d is

$$\begin{aligned} e_d &= \min_l \left(\|\mathbf{x}_{\tau^{(o)}} - \hat{\mathbf{x}}_{dl|\tau^{(o)}}\|_2^2 \right) \\ &= \min_l \left(\min_{\tau} \|\mathbf{x}_\tau - \mathbf{U}_{dl} \cdot (\mathbf{U}_{dl}^H \cdot \mathbf{x}_\tau)\|_2^2 \right) \end{aligned} \quad (18)$$

If

$$J = \arg \min_d e_d \quad (19)$$

then the algorithm decides that $\mathbf{x}$ belongs to target T_J .

In the above derivation, (17) shows that the optimal time-shift compensated approximation vector $\hat{\mathbf{x}}_{dl|\tau^{(o)}}$ has the minimum reconstruction error among the errors between $\mathbf{x}$ and its all approximation vectors $\hat{\mathbf{x}}_{dl|\tau}$, but it requires large computation burden to search the optimal time-shift compensation by shifting the test sample between range cells one by one. By formula derivation (as shown in the

appendix), we obtain

$$\begin{aligned} \hat{\mathbf{x}}_{dl|\tau^{(o)}} &= 1 - \arg \max_{\tau} \sum_{q=1}^p \left| \mathbf{u}_{dlq}^H \cdot \mathbf{x}_\tau \right|^2 \\ &= 1 - \arg \max_{\tau} \sum_{q=1}^p \left| \alpha_{dlq|\tau} \right|^2 \end{aligned} \quad (20)$$

where $\alpha_{dlq|\tau}$ can be regarded as not only the q th projection coefficient of $\mathbf{x}_\tau$ in the feature subspace $\mathbf{U}_{dl} = [\mathbf{u}_{dl1}, \mathbf{u}_{dl2}, \dots, \mathbf{u}_{dlq}, \dots, \mathbf{u}_{dlp}]$ (as shown in (15)), but also the correlation coefficient between $\mathbf{x}_\tau$ and $\mathbf{u}_{dlq}$. Therefore if we use the fast Fourier transformation, the computation burden can be extremely decreased. Thus the time-shift compensation between a test sample and a p -dimensional feature subspace requires the correlation operation for p times by (20), but the time-shift operation for N times by (17). Fig. 4 shows the flow chart of our recognition algorithm.

What discussed earlier is still suitable for real HRRPs, except that the transpose operator T should substitute for the complex conjugate transpose operator H . As stated in Section 2.1, the average profile (or the principal profile) of a real HRRP frame can be used as the corresponding frame template, thereby, the slide correlation processing between a test sample and the average profile can also be used as a time-shift compensation approach, which only requires the correlation operation once. Although the time-shift compensation obtained by this approach is not optimal, its computational burden is small. Actually, since the time-shift sensitivity of complex HRRPs is independent of their initial phases, the slide correlation processing can also be applied for complex HRRPs' time-shift compensation. In the following experiments, we will compare the two time-shift compensation approaches. In addition, as discussed in the literatures [7, 8], some data preprocessing methods, for example power transformation or Box-Cox transformation, can be used for real HRRPs to improve recognition performance. Since the PCA-based minimum reconstruction error approximation using the power transformed or Box-Cox transformed complex HRRPs is still independent of the initial phases, yet exploits the remaining phase

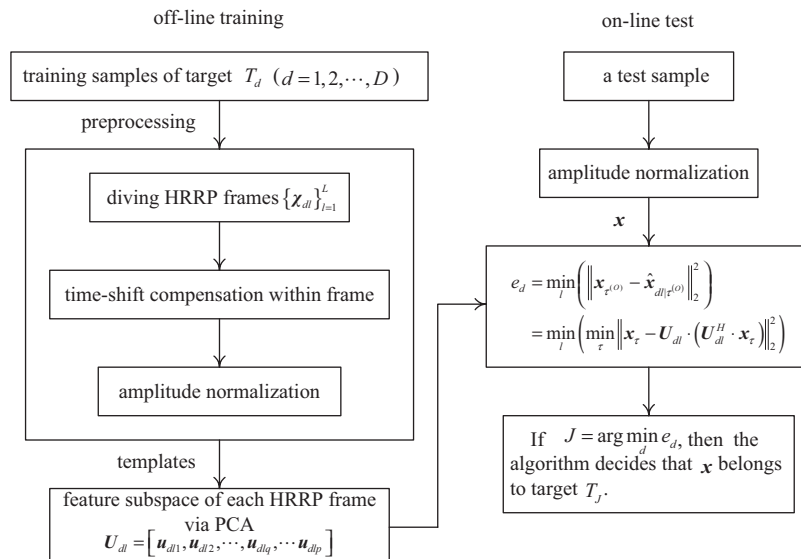
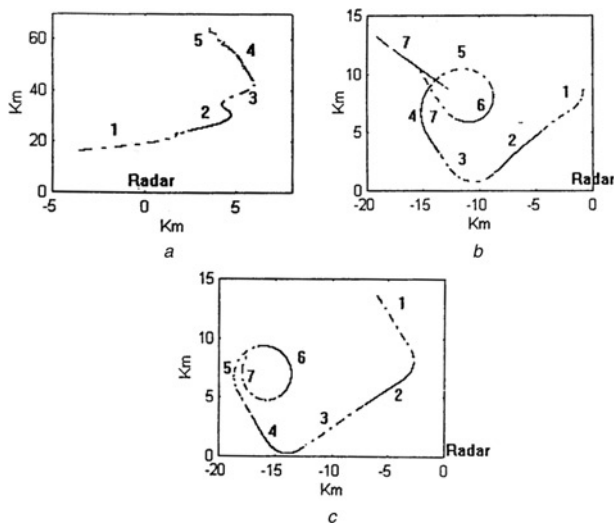


Fig. 4 Flow chart of the proposed recognition algorithm for complex HRRPs
PCA-based minimum reconstruction error approximation algorithm

Table 1: Parameters of planes and radar in the ISAR experiment

Radar parameters	Transmitted signal		Chirp signal
	waveform		
	Centre frequency, MHz		5520
	Bandwidth, MHz		400
	Pulse repetition frequency, Hz		400
	Sampling frequency after dechirp, MHz		10
Aircraft	Length, m	Width, m	Height, m
Yark-42	36.38	34.88	9.83
An-26	23.80	29.20	9.83
Cessna Citation S/II	14.40	15.90	4.57

**Fig. 5** Projections of target trajectories onto ground plane

Radar is at the grid origin

a Yark-42

b An-26

c Cessna Citation S/II

information in complex HRRPs, such data preprocessing methods can also be applied to complex HRRP-based RATR. The following experiments will show whether power transformation can improve the recognition rates of complex HRRPs or not.

4 Experimental result

4.1 Data description

The results presented in this section are based on measured HRRP data from flying airplanes. The parameters of the targets and radar are shown in Table 1, and the projections of target trajectories onto ground plane are shown in Fig. 5. The measured data are segmented, and every data segment has 26 000 samples. Two points should be considered when choosing training and test data. First, as mentioned earlier, target HRRP is very sensitive to target-aspect variation. Therefore the data used to train the classifier should cover all possible target-aspect theoretically. Secondly, since there is generally finite training data available to train the classifier, the generalisation ability is an important criterion to evaluate a classifier's performance. Therefore it is preferred to choose training and test data differently. From Fig. 5, we can approximately estimate the aspect angle of every target in every segment. According to the above two points, the second and the fifth segments of Yark-42, the fifth and the sixth segments of An-26, the sixth and the seventh segments of Cessna Citation S/ II are taken as the training samples, other data segments are taken as test samples. The elevation angles of the test data are different from those of the training data, thus the generalisation performance of the recognition methods can be tested. With large SNR, the effect of noise in the measured data on the recognition can be ignored. In addition, the measured HRRP is a 256-dimensional vector.

4.2 Recognition results

In our recognition experiments, the threshold η is set to be 99%. Tables 2 and 3 show the confusion matrices and average recognition rates of the PCA-based minimum reconstruction error approximation for original and optimally power transformed ($\alpha^* = 0.1$) real and complex HRRPs with test samples time-shift compensated by the optimal time-shift compensation and the slide correlation processing with the average profile of each real HRRP frame, respectively. By comparing the recognition results, first, the average recognition rates obtained by the optimal time-shift compensation are approximately 3% points larger than those obtained by the slide correlation processing, although the computation burden increases p times; secondly, with the same time-shift compensation approach, the average recognition rate for complex HRRPs is 4–7% points larger than that for real HRRPs; thirdly, by power

Table 2: Confusion matrices and average recognition rates for real HRRPs with test samples time-shift compensated by the optimal time-shift compensation and the slide correlation processing with the average profile of each real HRRP frame

Time-shift compensation approach	Original real HRRP			Original real HRRP			Real HRRP with optimal power transformation ($\alpha^* = 0.1$)		
	Slide correlation processing with the average profile			Optimal time-shift compensation			Optimal time-shift compensation		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	82.67	4.40	15.85	83.25	2.35	11.40	96.00	0.25	9.45
Cessna Citation S/II	0	80.10	1.50	0	84.50	0.90	1.00	95.35	3.60
An-26	17.33	15.50	82.65	16.75	13.15	87.70	3.00	4.40	86.95
Average recognition rate P_c , %	81.67			85.15			91.96		

Table 3: Confusion matrices and average recognition rates for complex HRRPs with test samples time-shift compensated by the optimal time-shift compensation and the slide correlation processing with the average profile of each real HRRP frame

Time-shift compensation approach	Original complex HRRP			Original complex HRRP			Complex HRRP with optimal power transformation ($\alpha^* = 0.1$)		
	Slide correlation processing with the average profile			Optimal time-shift compensation			Optimal time-shift compensation		
	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26	Yark-42	Cessna Citation S/II	An-26
Yark-42	89.92	1.25	10.85	89.83	0.40	6.15	99.75	0	5.25
Cessna Citation S/II	0	87.55	2.00	0	91.55	0.85	0	99.60	2.65
An-26	10.08	11.20	87.15	10.17	8.05	93.00	0.25	0.40	92.10
average recognition rate, P_c , %	88.21			91.46			96.50		

transformation with optimal parameter, the average recognition rates can be greatly improved not only for real HRRPs but also for complex HRRPs. Therefore the PCA-based minimum reconstruction error approximation is suitable for complex HRRP-based RATR and the phase information of complex HRRPs can be properly utilised in RATR for further improving the recognition performance.

5 Summary and discussion

A detailed study on the characteristics of complex HRRPs has been made, which includes the target-aspect, time-shift, amplitude-scale and initial phase sensitivity problems, the average complex profile and the principal complex profile of a complex HRRP frame and then points out how to deal with the initial phase sensitivity of complex HRRPs is a challenging topic in RATR using complex HRRPs. Since in the training phase the initial phases of a complex HRRP frame can be counteracted in the corresponding correlation matrix by outer product operation and in the test phase those of test samples can also be counteracted in the reconstruction error by inner product operation, the PCA-based minimum reconstruction error approximation is completely independent of the initial phases, yet exploits the remaining phase information in complex HRRPs for complex HRRP-based RATR. Furthermore, to make the recognition method more practical, a fast time-shift compensation algorithm is proposed in this paper. In the recognition experiments based on measured data, the PCA-based minimum reconstruction error approximation using complex HRRPs achieves better recognition results than that using real HRRPs.

However, we cannot draw the conclusion that the recognition performances of complex HRRPs are always better than those of real HRRPs, because some feature extraction methods and classification algorithms developed for real HRRPs cannot be applied for complex HRRPs directly. Therefore it is our future task to propose other feature extraction methods, classification algorithms or preprocessing methods for complex HRRPs.

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8 Appendix

Let $\mathbf{x}_\tau$ denote an amplitude normalised complex HRRP $\mathbf{x}$'s time-shifted vector with time shift of τ , and $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_q, \dots, \mathbf{u}_p]$ denote the p -dimensional feature subspace, where $\mathbf{u}_q$ is the q th eigenvector of the training data's correlation matrix $\mathbf{R} = E\{\mathbf{x}_m \cdot \mathbf{x}_m^H\} \simeq (1/M) \sum_{m=0}^{M-1} \mathbf{x}_m \cdot \mathbf{x}_m^H$ and the corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq \dots \geq \lambda_N$. The approximation vector

of $\mathbf{x}$ with the optimal time-shift compensation in $\mathbf{U}$ is

$$\begin{aligned}
 \hat{\mathbf{x}}_{\tau^{(o)}} &= \arg \min_{\tau} e_{\tau} = \arg \min_{\tau} \|\mathbf{x}_{\tau} - \mathbf{U} \cdot (\mathbf{U}^H \cdot \mathbf{x}_{\tau})\|_2^2 \\
 &= \arg \min_{\tau} \left[\|\mathbf{x}_{\tau}\|_2^2 - 2\mathbf{x}_{\tau}^H \cdot (\mathbf{U} \cdot \mathbf{U}^H) \cdot \mathbf{x}_{\tau} \right. \\
 &\quad \left. + \mathbf{x}_{\tau}^H \cdot \mathbf{U} \cdot \mathbf{U}^H \cdot \mathbf{U} \cdot \mathbf{U}^H \cdot \mathbf{x}_{\tau} \right] \\
 &= \arg \min_{\tau} [1 - 2\mathbf{x}_{\tau}^H \cdot (\mathbf{U} \cdot \mathbf{U}^H) \cdot \mathbf{x}_{\tau} \\
 &\quad + \mathbf{x}_{\tau}^H \cdot (\mathbf{U} \cdot \mathbf{U}^H) \cdot \mathbf{x}_{\tau}] \\
 &= 1 - \arg \max_{\tau} [\mathbf{x}_{\tau}^H \cdot (\mathbf{U} \cdot \mathbf{U}^H) \cdot \mathbf{x}_{\tau}] \\
 &= 1 - \arg \max_{\tau} \|\mathbf{U}^H \cdot \mathbf{x}_{\tau}\|^2 \\
 &= 1 - \arg \max_{\tau} \sum_{q=1}^p |\mathbf{u}_q^H \cdot \mathbf{x}_{\tau}|^2
 \end{aligned} \tag{21}$$